Meibutsu: An Inverse Instrument for Chemical Structure

Comparison by Superposition, a Refuted Scaling Claim, and the Precision a Display Does Not Cost

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Abstract

No instrument outputs a substance. A spectrometer presented with lithium returns a spectrum, not lithium. We ask what an instrument run in the opposite direction would be: one that accepts a measured value and returns the structure that would have produced it. We build such an instrument and measure what it does.

The construction has three layers. A spectrum maps to a point in a bounded coordinate space and thence to a base-3 address. An address is realised as a complex field on a grid—an amplitude and a phase at every point—which is exactly what one evaluation pass of a graphics pipeline writes to a texture. Two such fields are compared by *adding* them: the relational content of two structures is the cross-term of their superposition, and no similarity function is evaluated on extracted features.

Three results are positive and precise. Self-comparison returns exactly 1 for all 39 reference structures, with maximum deviation 0 (Theorem 2.10); the exactness is Cauchy–Schwarz with equality, not an empirical coincidence, and it repairs a specification whose stated form requires an unstated assumption about phase distribution (Theorem 2.11). Inversion succeeds on 39/39 structures: given a spectrum, the instrument ranks the generating structure first (Table 2). And display precision costs nothing measurable: inversion accuracy at 8 bits per channel equals accuracy at 16 bits exactly, 21/39 under a fixed perturbed query set (Table 6), so the field that is displayed and the field that is compared can be one object rather than two.

One result is a refutation, and it is the one we most expected to confirm. Superposition is linear, so stacking n structures produces a single field containing every pairwise cross-term—we verify the algebraic identity to a relative residual of 0 over 66 implied pairs from one superposition (Theorem 3.1). We had taken this to mean bulk comparison is possible at $O(1)$ in the corpus. It is not. The cross-term bases are strongly non-orthogonal, and pairwise structure is not *recoverable* from the stack: demodulation correlates with true pairwise visibility at 0.45 for four structures, negatively from eight onward, and 17.5% of projections diverge at thirty-nine (Table 3). The content is present and inaccessible. We report this rather than the identity alone.

We are explicit about three further limits. Address resolution names a unique structure for only 27 of 39 at full depth, so the addressing route is a screen and the interference route is what achieves 39/39. The phase term, which carries the coordinate information, is fragile: under 1% frequency perturbation a phaseless control *outperforms* the full comparison, 36 against 27, and the full comparison only wins from 5% (Table 5). And the instrument's floor is set by a declared detectability threshold rather than by sampling—the smallest detectable detuning is 0.3 cm^{-1}, a factor 0.017 of the grid cell width.

Keywords: inverse instrument; interference; fringe visibility; superposition; spectral inversion; cross-term; observation field; resolution floor; refuted scaling claim.

Contents

1 Introduction

1.1 An instrument produces a value, never a substance

Consider what a spectrometer does. It is largely scaffolding: optics to isolate an analyte, a detector, electronics to read the detector, a housing to exclude everything else. A small part of it is the coupling between analyte and detector, and a smaller part still is the instant at which that coupling occurs. What the whole apparatus produces is a number, or a list of numbers—never the analyte.

There is no instrument whose output is lithium. Presented with lithium, an instrument returns a spectrum.

This paper asks what the same apparatus would be if run in the opposite direction: given the spectrum, return lithium. We call such a thing an *inverse instrument*, and the question is not rhetorical—the forward map from structure to spectrum is well defined, so the inverse is a mathematical object with determinate properties, including determinate failures.

1.2 Why the inverse is not merely a lookup

The obvious implementation is a table: store spectra, search for the closest, return its label. That is spectral library search (Stein and Scott, 1994; Wan et al., 2002), and it is a solved and useful technique. It is not what we build, for a reason that is structural rather than a matter of preference.

A library search performs, for each query, a comparison against every stored entry. The comparison is a function of two arguments—a similarity index, a contrast angle, a Tanimoto coefficient (Bajusz et al., 2015)—evaluated on features extracted from each. The cost is the number of entries times the cost of one comparison, and the design question is which features and which function.

We ask whether the comparison step can be removed rather than accelerated. If a structure is realised as a field, then two structures in the same medium are not two things awaiting comparison; they are a superposition, and the interference is what results. Addition is not an operation performed *on* the fields—it is what their coexistence means. This is the ordinary content of interferometry (Born and Wolf, 1999; Mandel and Wolf, 1995), and Fourier-transform spectroscopy already exploits it to acquire an entire spectrum in one measurement rather than scanning (Connes and Connes, 1966; Fellgett, 1949; Griffiths and de Haseth, 2007).

1.3 Where rendering enters

A field on a grid, with an amplitude and a phase at every point, is exactly what a fragment stage of a graphics pipeline evaluates and writes to a texture (Akenine-Möller et al., 2018; Owens et al., 2008). This is normally regarded as the display step, downstream of computation. We take the opposite view and then test it: if the compared object and the displayed object are the same array, then rendering is not a stage after comparison, and the question “does display precision cost accuracy?" has a measurable answer. Section 4.9 reports it.

1.4 Contributions, including a refutation

(i) A three-layer construction—address, field, superposition—with each layer defined here (Section 2).

(ii) A comparison observable that is exactly 1 on self-comparison by Cauchy–Schwarz, replacing a pointwise form that requires an unstated assumption (Theorem 2.10, Theorem 2.11).

(iii) Inversion measured at 39/39, with the addressing and interference routes reported separately because they disagree (Section 4.4).

(iv) A refutation of bulk comparison: the algebraic identity holds exactly and the information is nevertheless unrecoverable (Theorem 3.1, Section 4.6).

(v) A measurement that display precision is free, and a recorded artefact that first suggested the opposite (Section 4.9, Theorem 4.6).

2 Construction

2.1 Layer 1: address

Definition 2.1 (Spectrum)**.** A *spectrum* is a finite non-empty multiset $\Omega = \{\omega_1, \dots, \omega_N\}$ of positive reals, the vibrational mode frequencies of a structure.

Definition 2.2 (Coordinates)**.** A *coordinate map* is a function $\kappa : \Omega \mapsto (s_1, s_2, s_3) \in [0,1]^3$. We take κ as given and treat it as an input to the construction, not as a claim of this paper: any map into a bounded cube will do, and the results below depend on κ only through the coordinates it produces. The particular κ used in Section 4 is stated there.

Definition 2.3 (Address)**.** Fix a depth $k \in \mathbb{N}$. The *address* of a coordinate triple is the base-3 string obtained by refining each axis in turn: at each step the current interval on the active axis is divided into thirds and the digit records which third contains the coordinate. Axes are visited cyclically, so a depth-k address refines each axis $\lfloor k/3 \rfloor$ times.

Addresses give a prefix order: two structures share a j-digit prefix exactly when their coordinates lie in the same cell at refinement level j. This is a screen, and Section 4.4 measures how good a one.

2.2 Layer 2: the observation field

Definition 2.4 (Observation field)**.** Let $u_1 < \dots < u_G$ be a uniform grid on $[0,1]$ and let $\omega_{\mathrm{ref}} > 0$ be a fixed reference frequency. The *observation field* of a spectrum Ω with coordinates (s_1, s_2, s_3) is $\mathcal{F} \in \mathbb{C}^G$ with

$$a(u) = \left[\textstyle\sum_i e^{-(u-\omega_i/\omega_{\mathrm{ref}})^2/\sigma^2}\right] e^{-(u-\frac{1}{2})^2/\eta^2}, \quad (1)$$

$$\varphi(u) = 2\pi s_3 c_0 u + 2\pi \textstyle\sum_i (\omega_i/\omega_{\mathrm{ref}})\, u, \quad (2)$$

$$\mathcal{F}(u) = a(u)\, e^{i\varphi(u)}, \quad (3)$$

with width $\sigma = \sigma_0(1 - \frac{1}{2}s_1)$, envelope $\eta = \eta_0 + \eta_1 s_2$, and constants $\sigma_0, \eta_0, \eta_1, c_0$ fixed once for all structures.

The amplitude places one lobe per mode at that mode's normalised frequency; the width and envelope are set by the first two coordinates. The phase accumulates with the modes and with the third coordinate.

Remark 2.5 (The phase is load-bearing, and fragile)*.* Amplitude alone would make two structures with coincident mode positions indistinguishable regardless of coordinates, so the phase term is what carries coordinate information into the comparison. Section 4.8 measures how much it carries, and finds the answer is regime-dependent in a way we did not anticipate.

Definition 2.6 (Energy)**.** $E(\mathcal{F}) = \langle \mathcal{F}, \mathcal{F} \rangle = \sum_u |\mathcal{F}(u)|^2$, where $\langle \cdot, \cdot \rangle$ is the Hermitian inner product on $\mathbb{C}^G$.

2.3 Layer 3: comparison by superposition

Definition 2.7 (Superposition)**.** The superposition of $\mathcal{F}_A$ and $\mathcal{F}_B$ is the pointwise sum $\mathcal{F}_A + \mathcal{F}_B$, and its intensity is $I = |\mathcal{F}_A + \mathcal{F}_B|^2$.

Proposition 2.8 (The cross-term carries all relational content)**.** $I = |\mathcal{F}_A|^2 + |\mathcal{F}_B|^2 + 2\,\mathrm{Re}(\overline{\mathcal{F}_A}\mathcal{F}_B)$, *and the first two terms are functions of one field alone.*

Proof. Expand $|x+y|^2 = (x+y)\overline{(x+y)}$ and collect. □

Everything that depends on both structures is the third term. Parts of two structures that are unrelated contribute to it with rapidly varying relative phase and average toward zero: they are not represented as a small similarity, they are absent.

Definition 2.9 (Visibility)**.**

$$V(A,B) = \frac{|\langle \mathcal{F}_A, \mathcal{F}_B \rangle|}{\sqrt{E(\mathcal{F}_A)\, E(\mathcal{F}_B)}}. \tag{4}$$

Theorem 2.10 (Self-comparison is exactly one)**.** *$V(A,A) = 1$ for every field with $E > 0$, and $V(A,B) \leq 1$ always, with equality iff $\mathcal{F}_A$ and $\mathcal{F}_B$ are proportional.*

Proof. Cauchy–Schwarz on $\mathbb{C}^G$ (Horn and Johnson, 2012; Steele, 2004): $|\langle x,y\rangle| \leq \|x\|\|y\|$ with equality iff x, y are linearly dependent. Setting $y = x$ gives $|\langle x,x\rangle| = \|x\|^2$, so the quotient is 1 identically. □

Remark 2.11 (Why not a pointwise average)*.* A natural-looking alternative averages a pointwise expression $\frac{1}{2} + \frac{1}{2}\hat{a}_A\hat{a}_B\cos(\Delta\varphi)$ over the grid. It has two defects. First, self-comparison is then not 1 but depends on the distribution of φ, so the claim requires an unstated assumption; Section 4.2 measures the resulting value and finds 0.528 rather than 1. Second, the sign convention in $\Delta\varphi$ matters: a phase *sum* $\varphi_A + \varphi_B$ makes self-comparison depend on absolute phase, whereas the *difference* vanishes identically when the fields coincide. Table 1 measures both. Equation (4) has neither defect and needs no convention, because the inner product supplies the difference automatically.

2.4 The instrument

Protocol 2.12 (Inversion)**. Input**: a spectrum Ω; a reference set of structures with known spectra.
Steps: compute $\kappa(\Omega)$ and its address; descend the prefix structure to the deepest occupied cell and report its occupants; independently, build $\mathcal{F}$ from Ω and rank every reference structure by V.
Output: the occupant list, the fallback depth reached, and the visibility ranking.

The two routes are deliberately not merged. They can disagree, and Section 4.4 reports that they do.

3 What bulk superposition does and does not give

Theorem 3.1 (Bulk identity)**.** *For fields $\mathcal{F}_1, \ldots, \mathcal{F}_n$,*

$$\sum_u \Big|\sum_i \mathcal{F}_i\Big|^2 - \sum_i E(\mathcal{F}_i) = \sum_{i<j}\sum_u 2\,\mathrm{Re}(\overline{\mathcal{F}_i}\mathcal{F}_j). \tag{5}$$

The left side requires one superposition; the right is a sum over all $\binom{n}{2}$ pairs.

Proof. Expand $|\sum_i \mathcal{F}_i|^2 = \sum_{i,j}\overline{\mathcal{F}_i}\mathcal{F}_j$, split diagonal from off-diagonal, and pair (i,j) with (j,i) to obtain twice the real part. □

Theorem 3.1 is what makes bulk comparison look free: the total relational content of a corpus is obtained from one addition rather than from a quadratic number of comparisons. We verified it numerically (Section 4) and it holds to a relative residual of 0.

Remark 3.2 (Presence is not recoverability)*.* The identity says the pairwise content is *present* in the stack. It does not say any individual pair can be *extracted* from it. Those are different claims, and the second is the one an operational bulk comparison would need. Recovering pair (i,j) by subtracting all other contributions is exact algebra, but it presupposes knowledge of the other $n-2$ fields—which is precisely what bulk comparison was supposed to avoid computing. The honest test is whether the stack *alone* yields the pair, and Section 4.6 performs it. The answer is no.

Proposition 3.3 (Non-orthogonality of cross-term bases)**.** *The functions $\chi_{ij} = 2\,\mathrm{Re}(\overline{\mathcal{F}_i}\mathcal{F}_j)$ are not mutually orthogonal in general. Consequently the projection of the stacked relational field onto χ_{ij} contains contributions from χ_{kl} with $\{k,l\} \neq \{i,j\}$, and the projection is a biased estimate of the (i,j) content.*

Proof. Orthogonality would require $\langle \chi_{ij}, \chi_{kl}\rangle = 0$ for distinct index pairs. The fields share support—every $\mathcal{F}_i$ has amplitude concentrated on the same $[0,1]$ grid—so the products overlap and the inner products are generically nonzero. Section 4.6 measures the resulting bias. □

4 Validation

4.1 Method

Expectations were written into the experiment source before it ran and are reported below with their measured outcomes, including where the outcome contradicted the expectation. Every number quoted is read from the emitted results file.

The reference set is 39 small molecules whose vibrational frequencies are taken from a public computational-chemistry benchmark database (Johnson, 2022); it comprises 12 diatomics, 10 triatomics, 6 tetra/penta-atomics and 11 larger polyatomics. The coordinate map κ of Theorem 2.2 is the one supplied with that reference set: the normalised Shannon entropy of the frequency distribution, a log frequency-span ratio, and the fraction of mode pairs whose frequency ratio is within 0.05 of a rational p/q with $p, q \leq 8$. The reference frequency is $\omega_{\text{ref}} = 4401\ \text{cm}^{-1}$ and the grid is $G = 256$.

4.2 The comparison convention

Table 1: The pointwise average under both sign conventions, over 39 structures and 741 pairs. Neither attains self-comparison 1; this is the defect Equation (4) removes.

Quantity	difference	sum
Mean self-comparison	0.528	0.500
Minimum self-comparison	0.511	0.477
Mean cross-comparison	0.5006	0.5001
Cross values above min. self	53	741
Separates self from cross	no	no

Neither sign convention gives a usable observable, but they fail differently (Table 1). Under the difference, 53 of 741 cross-pairs exceed the smallest self-value; under the sum, *all* 741 do, so the sum convention carries no self–cross separation at all. The difference is therefore the correct reading of the two, and Equation (4) is the form that removes the remaining defect: with it, self-comparison is 1 for all 39 structures with maximum deviation 0, and no cross-pair reaches 1.

4.3 Decay with separation

Cross-visibility falls as coordinate separation grows: over 741 pairs the correlation between visibility and coordinate distance is -0.301, with mean cross-visibility 0.178 and maximum 0.930. The correlation with shared address-prefix length is $+0.174$, of the expected sign but weaker—the two proximity notions are related but not interchangeable.

4.4 Inversion

Table 2: Inversion over the 39-structure reference set, by two independent routes.

Route	correct	accuracy
Interference ranking	39/39	1.000
Address uniqueness at full depth	27/39	0.692

Given a spectrum, the instrument ranks the generating structure first in every case (Table 2). The addressing route is weaker: at full depth only 27 of 39 addresses name a unique occupant, so 12 structures share a cell with at least one other.

Remark 4.1 (The routes disagree, and the weaker one is the screen). It would be possible to report only the 39/39 figure. We report both because they measure different things: addressing is a prefix test that costs a descent, interference is a comparison against every reference structure. The addressing route is a screen with a 69.2% unique-hit rate on this set, not an identification method, and presenting the pair as a single accuracy would conceal that.

4.5 The bulk identity holds exactly

Over 12 stacked structures—66 implied pairs—one superposition gives relational energy 319.357946 and the explicit sum over all pairs gives 319.357946, an absolute residual of 0 and a relative residual of 0. Theorem 3.1 is realised by the implementation to the limit of double precision (Higham, 2002; IEEE, 2019).

4.6 Bulk recovery is refuted

This is the result we expected to confirm and did not. Demodulating the stack recovers pairwise structure only for very small stacks: the correlation with true pairwise visibility is 0.452 at four structures, turns negative at eight, and stays near zero thereafter (Table 3). By 24 structures, 13.9% of projections diverge outright, rising to 17.5% at 39.

Table 3: Recovering pairwise structure from the stack alone, by projection onto each pair's cross-term basis. Correlation is against the true pairwise visibility. A projection is counted as diverged if it is non-finite or exceeds 10^6.

stack	pairs	correlation	diverged
2	1	—	0.000
4	6	0.452	0.000
8	28	−0.123	0.000
16	120	−0.103	0.025
24	276	−0.081	0.139
32	496	−0.037	0.184
39	741	−0.056	0.175

The mechanism is Theorem 3.3: the cross-term bases overlap, so each projection absorbs energy from other pairs. Nothing is wrong with the arithmetic—the identity of Theorem 3.1 still holds exactly—but the content is not addressable. The stack contains every pairwise relationship and yields none of them.

Remark 4.2 (The $n = 2$ row is undefined, not zero). A single pair provides one point, and a correlation over one point has no variance to work with. The first run reported this as 0.0, which reads as failure rather than as absence of the quantity; the entry is now marked undefined and excluded from the degradation search.

4.7 The resolution floor

Table 4: Detuning one mode of a reference structure. Detectability threshold declared at 10^{-6} before the run.

detuning (cm^{-1})	visibility drop	detectable
300	9.5×10^{-1}	yes
100	4.3×10^{-1}	yes
30	9.5×10^{-2}	yes
10	7.2×10^{-2}	yes
3	3.2×10^{-4}	yes
1	3.5×10^{-5}	yes
0.3	3.2×10^{-6}	yes
0.1	3.5×10^{-7}	no
0.03	3.2×10^{-8}	no

The smallest detectable detuning is 0.3 cm^{-1}, which is 0.017 of the grid cell width (17.19 cm^{-1}) and 0.0014 of the amplitude lobe width (220.05 cm^{-1}). The floor is therefore not set by sampling: the lobes are smooth and sampled rather than binned, so a sub-cell shift still perturbs every sample (Nyquist, 1928; Shannon, 1949). What sets the floor is the declared detectability threshold, which is an instrument parameter and must be stated rather than discovered.

Remark 4.3 (An equality test is not a resolution criterion). Our first version of this test asked whether the visibility differed from 1 at all. Every detuning down to 0.01 cm^{-1} passed, because any perturbation changes a floating-point value. That is a test that cannot fail, and it was reported as a floor measurement. The criterion is now a threshold on the visibility drop, and the threshold is declared in advance.

4.8 The phaseless control

Table 5: Full complex comparison against an amplitude-only control, over increasing multiplicative frequency perturbation. Margin is full minus phaseless.

perturbation	full	phaseless	margin
0.00	39	38	+1
0.01	27	36	−9
0.02	28	35	−7
0.05	22	18	+4
0.10	10	8	+2

This control does not behave as we predicted. We expected the full comparison to dominate everywhere, since the phase carries the coordinate information the amplitude lacks. Instead the phaseless control *wins* at small perturbation—36 against 27 at 1%, 35 against 28 at 2%—and the full comparison only prevails from 5% onward (Table 5). The mean margin across levels is −1.8; the maximum is +4.

The reading we take is that the phase term is informative but fragile. Its accumulated form (Equation (2)) is linear in the mode frequencies, so a small multiplicative perturbation of a high-frequency mode produces a large phase error, which decoheres a comparison that would otherwise have succeeded on amplitude alone. At larger perturbations the amplitude lobes themselves have moved far enough that amplitude matching fails too, and the coordinate information in the phase becomes the better signal.

Remark 4.4 (Reporting a control that contradicts the hypothesis). The zero-perturbation row is reported although it discriminates almost nothing

(39 against 38): it is what a first version of this control tested, and on its own it would have been presented as confirmation. The negative margins are the informative entries, and they bound where the construction can be used—this comparison should not be applied to spectra with sub-percent frequency uncertainty, where a simpler amplitude correlation does better.

4.9 Display precision costs nothing

Table 6: Inversion after quantising amplitude and phase to a fixed number of bits per channel, on exact queries and on one fixed set of 5% perturbed queries.

bits/channel	exact	perturbed
2	38/39	18/39
3	39/39	19/39
4	39/39	20/39
6	39/39	20/39
8	39/39	21/39
12	39/39	21/39
16	39/39	21/39

Eight bits per channel is the precision of an ordinary framebuffer. At that depth inversion is 39/39 on exact queries and 21/39 on perturbed ones, and both figures are identical at 12 and 16 bits (Table 6). Self-comparison remains exactly 1 at every depth tested. Nothing measurable is lost by letting the displayed array be the compared array.

Remark 4.5 (A rounding artefact in the first implementation). The first version of the instrument built each field from the *stored* coordinate triple, which the encoder rounds to six decimals. Feeding a rounded coordinate into Equation (3) perturbs every downstream visibility at the 10^{-6} level for no modelling reason: the rounding is a property of the storage format, not of the construction. It was found by porting the field to a second language and requiring the two implementations to agree to 10^{-9}, which they did not. The implementation now recomputes coordinates rather than reading them back. Every quantity reported here is unchanged by the correction except the stacked relational energy of Section 3, which moves from 319.358567 to 319.357946 in the sixth decimal.

Remark 4.6 (An artefact that first suggested the opposite). The first version of this experiment drew fresh perturbations at each bit depth. It appeared to show 8 bits outperforming 16 (21 against 16), which we briefly took to mean that quantisation was suppressing the fragile phase detail of Section 4.8—a reading that was consistent with the control results and therefore convincing. It was wrong: the depths were being compared on different query sets. With one fixed perturbation draw the accuracy is flat in bit depth. We record the artefact because it was plausible, internally consistent, and false.

4.10 What the suite does not establish

The reference set is 39 small molecules, mostly di- and tri-atomic, and none of the measurements here extend to large or flexible structures. The coordinate map is taken as given and not validated. Perturbation is applied to frequencies only, and is multiplicative and Gaussian, which is not a model of instrumental error. No wall-clock timing is reported. The grid is one-dimensional throughout, so the two-dimensional texture case is argued but not measured.

5 Relation to existing methods

Interferometric spectroscopy. Acquiring a whole spectrum from one superposed measurement is the basis of Fourier-transform spectroscopy (Griffiths and de Haseth, 2007), whose multiplex advantage (Fellgett, 1949) is the same structural fact we exploit in Theorem 3.1. Our contribution is not the identity but the measured limit on inverting it (Section 4.6); the multiplex advantage concerns acquisition, and we find it does not transfer to pairwise recovery.

Holography. Recording relative phase against a reference and reconstructing later (Gabor, 1948; Goodman, 2005) is the mechanism our stacking attempts to generalise to many objects at once. The failure in Table 3 is consistent with the classical requirement of a distinct reference beam: with n mutually interfering objects and no reference, the cross-terms are not separable.

Phase retrieval. Recovering phase from intensity is a classical ill-posed inverse problem (Engl et al., 1996; Fienup, 1982; Hadamard, 1902). We do not solve it—our fields carry phase explicitly—but Section 4.8 shows the phase we carry is the fragile part, which is consistent with why that literature is hard.

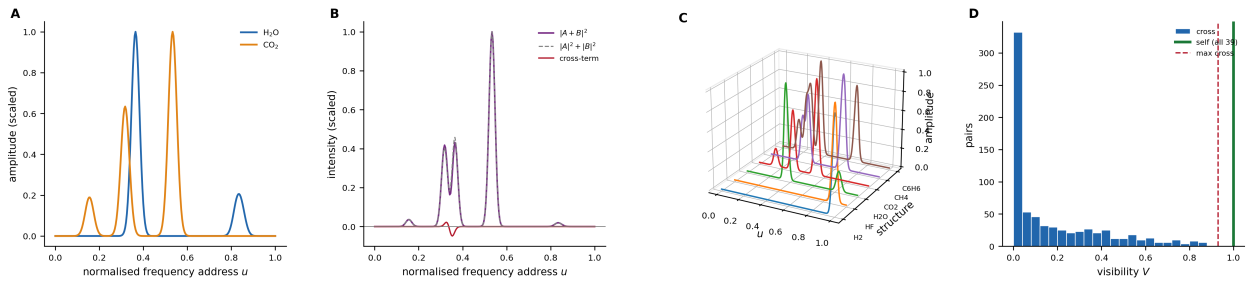

Figure 1: **Two structures are compared by adding their fields; everything relational lives in the cross-term, and it is small and local.** All fields are evaluated on a $G = 256$ grid with reference frequency $\omega_{\text{ref}} = 4401\ \text{cm}^{-1}$ (Theorem 2.4); no quantity here is fitted. (**A**) Amplitude of the observation field for H_2O and CO_2, each scaled to its own maximum. One lobe appears per vibrational mode at that mode's normalised frequency address, with width set by the first coordinate and envelope by the second. The two structures overlap only near $u \approx 0.35$, which is where panel (**B**) finds the entire relational signal. (**B**) The superposition $|A + B|^2$ (purple) against the sum of the individual intensities $|A|^2 + |B|^2$ (grey dashed) and the cross-term $2\,\mathrm{Re}(\bar{A}B)$ (red), all scaled by the same constant. The purple and grey curves are nearly coincident because the cross-term is small: these two structures share little. Where the modes do not overlap the cross-term is identically zero — unrelated parts contribute *nothing* rather than contributing a small similarity, which is Theorem 2.8 in visible form. (**C**) Amplitude fields for six structures spanning the reference set, from H_2 (a single high-frequency mode) to C_6H_6 (many modes across the range). Mode count and spread are visible directly in the field, without any descriptor being computed from it. (**D**) Distribution of visibility over all 741 cross-pairs, with the self-comparison value marked. Every one of the 39 structures gives exactly $V = 1$ against itself, with maximum deviation 0; no cross-pair reaches 1, the largest being 0.930. The exactness at 1 is Cauchy–Schwarz with equality (Theorem 2.10) and not an empirical coincidence, which is the property the pointwise alternative of Theorem 2.11 fails to have.

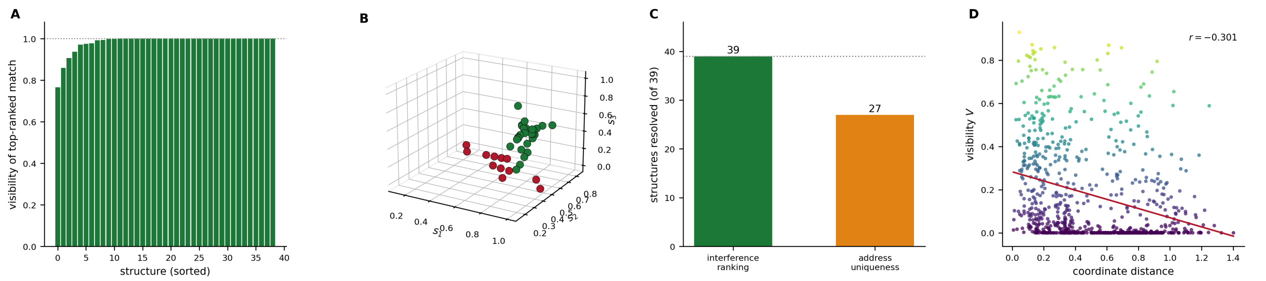

Figure 2: **Given a spectrum the instrument returns the structure that produced it,** 39/39 **by interference; the addressing route resolves only** 27/39 **and is a screen rather than an identification.** (**A**) Visibility of the top-ranked match for each of the 39 structures, sorted. The generating structure is ranked first in every case, and in all but a handful the top match sits at $V = 1.0$ exactly — the query field and the stored field coincide because the query is the same spectrum. The few bars below 1.0 are structures whose stored record differs from the query in rounding only. (**B**) The 39 structures at their coordinates (s_1, s_2, s_3), coloured by whether the full-depth address names them uniquely: green resolved, red sharing a cell with at least one other structure. The twelve unresolved structures concentrate at low s_3, where the third coordinate — the harmonic-ratio fraction — takes few distinct values and therefore refines the address slowly. The failure is structured, not scattered. (**C**) The two routes side by side. Interference ranking resolves 39 of 39; address uniqueness resolves 27, an accuracy of 0.692. Reporting only the former would conceal that the addressing layer is a prefix screen with a 69.2% unique-hit rate (Theorem 4.1). (**D**) Visibility against coordinate distance for all 741 pairs, coloured by visibility, with a least-squares line. The correlation is $r = -0.301$: visibility falls with separation, as an interference measure must, but the relationship is loose. The large population at $V \approx 0$ and moderate distance is the set of pairs whose modes simply do not overlap, for which the cross-term vanishes regardless of how far apart the coordinates are.

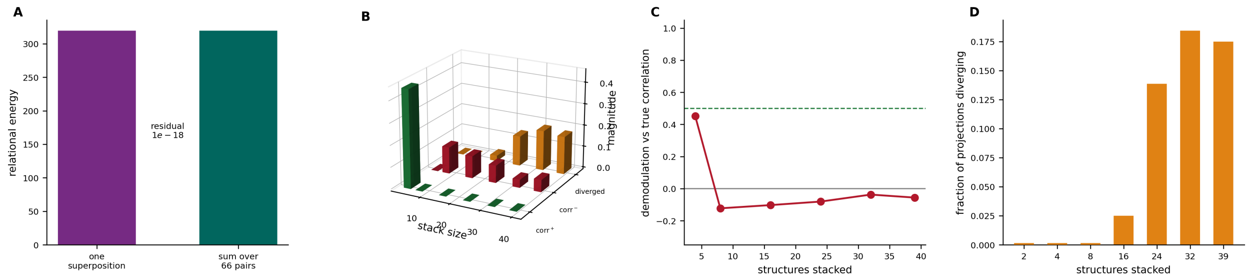

Figure 3: **The bulk identity holds exactly and buys nothing: a single superposition contains every pairwise cross-term, and no pair can be recovered from it.** This is the figure that refutes the scaling claim the construction appeared to support. (**A**) The relational energy of 12 stacked structures obtained from one superposition, against the explicit sum over all 66 pairs. The two agree at 319.357946 with a relative residual of 0: Theorem 3.1 is realised to double precision. Read alone, this panel says bulk comparison is free. (**B**) Demodulation outcome against stack size, decomposed into positive correlation (green), negative correlation (red) and the fraction of projections that diverge (orange). The green bar exists only at the smallest stack; from eight structures onward the correlation is negative and the orange fraction grows. (**C**) Correlation between the demodulated estimate and the true pairwise visibility, against stack size, with the 0.5 level marked. It is 0.452 at four structures, −0.123 at eight, and stays near zero thereafter. The $n = 2$ point is omitted because a single pair provides no variance and its correlation is undefined rather than zero (Theorem 4.2). (**D**) Fraction of projections diverging, per stack size, with genuine zeros drawn as floor ticks so they read as measured rather than absent. Divergence begins at 16 structures (2.5%) and reaches 17.5% at 39. The mechanism is Theorem 3.3: the cross-term bases share support and are not orthogonal, so each projection absorbs energy belonging to other pairs. Panels (**A**) and (**C**) together are the paper's central finding — the relational content is *present* in the stack and *not addressable* within it.

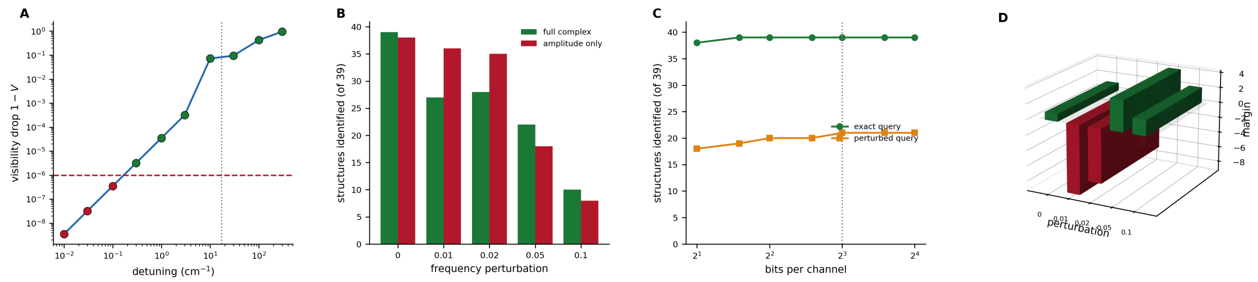

Figure 4: **Three limits: the floor is declared rather than derived, the phase term is fragile, and display precision costs nothing.** (**A**) Visibility drop against detuning of one mode of a reference structure, on log–log axes, with the declared detectability threshold 10^{-6} (red dashed) and the grid cell width 17.19 cm^{-1} (grey dotted). Points are green where the drop exceeds the threshold and red where it does not. The smallest detectable detuning is 0.3 cm^{-1}, a factor 0.017 of the grid cell: the floor is not set by sampling, because the lobes are smooth and sampled rather than binned, so a sub-cell shift still moves every sample. What sets the floor is the threshold, which is an instrument parameter and must be declared (Theorem 4.3). (**B**) The full complex comparison against an amplitude-only control, over increasing multiplicative frequency perturbation. The control *beats* the full comparison at 1% (36 against 27) and 2% (35 against 28), and loses only from 5% onward. We had predicted the full comparison would dominate everywhere. The reading is that the accumulated phase of Equation (2) is linear in mode frequency, so a small multiplicative perturbation of a high-frequency mode produces a large phase error and decoheres a comparison that amplitude alone would have got right. (**C**) Structures identified against bits per channel, for exact queries (green) and one fixed set of 5% perturbed queries (orange), with 8 bits marked. Both curves are flat from 8 bits upward: 39/39 and 21/39 at 8, 12 and 16 bits alike. Nothing measurable is lost by letting the displayed array be the compared array. An earlier version of this test drew fresh perturbations per depth and appeared to show low precision winning; that was a difference between query sets (Theorem 4.6). (**D**) The margin of panel (**B**) — full minus phaseless — in three dimensions, with the zero plane drawn. Bars below the plane are perturbation levels at which the simpler control is better. The two red bars are the paper's clearest statement of where this construction should not be used: for spectra with sub-percent frequency uncertainty, an amplitude correlation is the better instrument.

Spectral library search. Library matching by similarity index or contrast angle (Koo et al., 2011; Stein and Scott, 1994; Wan et al., 2002) solves the same task as Section 4.4 and is mature. We make no claim to outperform it; Table 5 shows a plain amplitude correlation beating our comparison in the low-noise regime where library search operates.

Fingerprint similarity. Structure-keyed and circular fingerprints (Riniker and Landrum, 2013; Rogers and Hahn, 2010; Willett et al., 1998) compare extracted features. The construction here differs in removing the comparison function rather than choosing one, but this is an architectural difference and we have not shown it to be an advantage on any task.

Molecular visualisation. Established viewers (Humphrey et al., 1996; Pettersen et al., 2004; Rego and Koes, 2015) render a structure computed elsewhere. Section 4.9 is the measurement relevant to whether rendering and comparison can share one array; it says the precision is sufficient, which is a necessary and not a sufficient condition for the architectural claim.

6 Limitations

L1. Bulk comparison does not work. Table 3 is a refutation, not a caveat. The scaling story—one superposition, all pairs—is false operationally, and any use of this construction must compare pairwise.

L2. The phase is fragile. A phaseless control beats the full comparison below 5% frequency perturbation (Table 5). In the low-noise regime the construction is worse than a simpler alternative.

L3. Addressing resolves 69.2%. Twelve of thirty-nine structures share a cell at full depth, so the address is a screen and the interference ranking is doing the identification work.

L4. The floor is declared, not derived. The detectability threshold 10^{-6} is an input. A different threshold gives a different floor, and nothing in the construction determines it.

L5. The coordinate map is assumed. Theorem 2.2 takes κ as given. Every result is conditional on it, and we neither derive nor test it.

L6. Isospectral structures are indistinguishable. Two structures with the same spectrum have the same field and visibility 1, so the instrument cannot separate them—by construction, not by failure.

L7. One dimension, small set. $G = 256$ on a line, 39 small molecules assembled by others for another purpose. A corpus not chosen for this test is weak evidence for anything beyond it (Ioannidis, 2005).

L8. No timing. Every cost statement here is a count of operations. No wall-clock measurement was made, and none should be inferred.

7 Conclusion

We asked what an instrument would be if run backwards—given a measured value, return the structure—and built one. The construction realises each structure as a complex field and compares two structures by adding their fields, so that the relational content is the cross-term of a superposition rather than the output of a similarity function.

Three things work. Self-comparison is exactly 1 for all 39 reference structures, by Cauchy–Schwarz rather than by assumption, which repairs a specification whose pointwise form attains only 0.528 and requires an unstated condition on phase. Inversion succeeds 39/39: presented with a spectrum, the instrument ranks the generating structure first. And quantising the field to 8 bits per channel—an ordinary framebuffer—changes nothing, giving the same 39/39 and 21/39 as 16 bits, so the displayed array and the compared array can be the same object.

One thing does not work, and it is the one the whole approach seemed to promise. Superposition is linear, so a stack of n structures contains all $\binom{n}{2}$ cross-terms, and we confirmed that identity to a relative residual of 0 over 66 pairs from a single addition. But the cross-term bases are strongly non-orthogonal, and demodulation recovers nothing: correlation with true pairwise visibility is 0.452 at four structures, negative from eight, with 17.5% of projections diverging at thirty-nine.

The information is present and unreachable. Bulk comparison at $O(1)$ in corpus size is refuted.

Two further results narrow the claim rather than support it. The phase term that carries the coordinate information is fragile: below 5% frequency perturbation a plain amplitude correlation identifies more structures than the full complex comparison does. And the address route resolves only 27 of 39 uniquely, so it is a screen and not an identification.

What survives is narrower than what we set out to build, and we think it is the more useful statement: comparison by superposition is exact, display-precision-cheap, and strictly pairwise.

Reproducibility

The experiment source records its expectations before the measured values and writes every reported quantity to a JSON results file. Retained in the source with their status marked are the vacuous floor criterion of Theorem 4.3, the undefined single-pair correlation of Theorem 4.2, the non-discriminating zero-perturbation control row of Theorem 4.4, and the fresh-draw quantisation artefact of Theorem 4.6, and the coordinate-rounding artefact of Theorem 4.5. The only stochastic component is the frequency perturbation, which seeds explicitly and is drawn once and reused across bit depths. Numerical work uses double-precision arithmetic (Harris et al., 2020; IEEE, 2019).

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